

DSC 335, Handout 15: Naive Bayes Classification (*Chapter 14*)

LEARNING OBJECTIVES.

- *Build a Bayesian regression model and simulate its posterior.*
- *Implement and evaluate Bayesian classification models.*

Logistic regression allows us to classify individuals in terms of a binary response. But what if we want to classify individuals in terms of a categorical response (with an arbitrary number of categories)?

Naive Bayes Classification

Let Y denote a categorical response variable with two or more categories and $(X_1, \dots, X_p)$ be a set of p possible predictors of Y . Then the posterior probability that a new case with observed predictors $(X_1, \dots, X_p) = (x_1, \dots, x_p)$ belongs to class $Y = y$ is

In order to simplify the above calculation, Naive Bayes classification makes the following naive assumptions about the likelihood function of Y :

- Predictors X_i are conditionally independent, thus
- For categorical predictors X_i , the conditional pmf

is estimated by the sample proportion of $Y = y$ cases for which $X_i = x_i$.

- For quantitative predictors X_i , the conditional pmf/pdf $f(x_i|y)$ is defined by a Normal model:

That is, within each category $Y = y$, we assume that X_i is Normally distributed around some mean μ_{iy} with standard deviation σ_{iy} . Further, we estimate μ_{iy} and σ_{iy} by the sample mean and standard deviation of the observed X_i values in category Y_i .

Example 1. We are interested in classifying Centre students as liberal, moderate, or conservative based on our class survey data. We will consider the **athlete** and **absence** variables and as an example classify a student who is not an athlete and had four absences.

- (a) Create and summarize a graphical display of the relationship between **athlete** and **politics**.
- (b) Using only **athlete**, implement a Naive Bayes classifier. What would we classify our example student as?
- (c) Create and summarize a graphical display of the relationship between **absence** and **politics**.
- (d) Using only **absence**, implement a Naive Bayes classifier. What would we classify our example student as?
- (e) Using both **athlete** and **absence**, implement a Naive Bayes classifier. What would we classify our example student as?

Evaluating Naive Bayes Classification

Example 2. Evaluate the three classification models above using both in-sample and 5-fold cross-validation procedures and summarize the results. What could we do to improve the model accuracy?

Example 3. Use Naive Bayes as an alternative to the logistic regression model in Handout 14, Example 5. How do the in-sample and 4-fold cross-validation metrics compare?